C-GenReg: Training-Free 3D Point Cloud Registration by Multi-View-Consistent Geometry-to-Image Generation with Probabilistic Modalities Fusion

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https://arxiv.org/abs/2604.16680

Research Questions

- How can World Foundation Models generate multi-view consistent RGB images from 3D point clouds to enable zero-shot registration?
- How can geometric and generated RGB correspondences be probabilistically fused without training to improve registration robustness?
- Can generative registration frameworks work on real outdoor LiDAR data without any RGB imagery?

Core Idea

Point cloud registration struggles with cross-domain generalization, but vision foundation models (VFMs) excel at image matching. C-GenReg transfers the 3D registration problem to the image domain by using a World Foundation Model to generate multi-view consistent RGB views from depth maps. A task-specific VFM then extracts correspondences from these synthetic images, which are lifted back to 3D. The key insight is combining these generated-RGB correspondences with traditional geometric correspondences through a probabilistic 'Match-then-Fuse' scheme that treats each modality as independent evidence sources, preserving their respective inductive biases while providing calibrated confidence estimates.

Key Formula

Noisy-AND Probabilistic Fusion

$$p_{ij}^{fuse} = \frac{p_{ij}^{img} p_{ij}^{geo} (1-\pi_{ij})}{p_{ij}^{img} p_{ij}^{geo} (1-\pi_{ij}) + (1-p_{ij}^{img})(1-p_{ij}^{geo})\pi_{ij}}$$

p_{ij}^{img} and p_{ij}^{geo} are correspondence posteriors from image and geometric branches, π_{ij} is prior matching probability, p_{ij}^{fuse} is fused posterior. Combines independent modality evidence under conditional independence assumption to favor correspondences supported by both modalities.

Noisy-OR Probabilistic Fusion

$$p_{ij}^{Noisy-OR} = 1 - (1-p_{ij}^{img})(1-p_{ij}^{geo})$$

Alternative fusion that aggregates evidence in union-like manner, increasing confidence when either image or geometric branch provides strong support. Used for comparison with the main Noisy-AND approach.

Key Figure

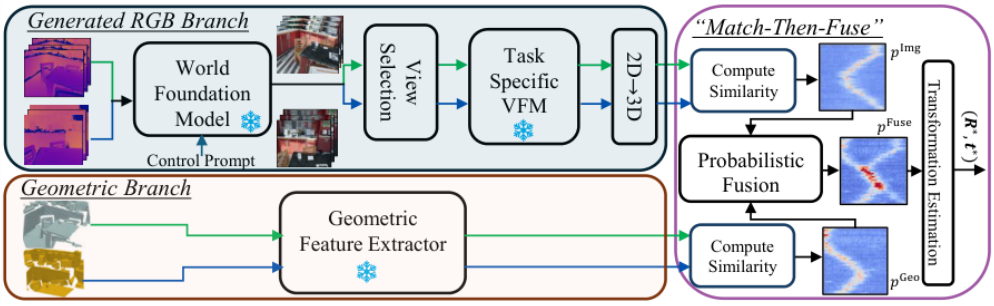

Figure 2. ***C-GenReg* Overview**: A training-free, zero-shot point cloud registration framework with two parallel branches. (1) **Generated-RGB Branch** - **source** and **target** point clouds are each represented as depth-frame sequences, temporally concatenated and processed by a frozen *World Foundation Model* to generate RGB views that are geometrically aligned and appearance-consistent across views. A subset of K frames per domain is fed to a frozen, task-specific *Vision Foundation Model* (VFM) to extract dense pixel-level features, later lifted to 3D using the original depths. (2) **Geometric Branch** - extracts dense geometric features directly from the raw point clouds using a pretrained geometric feature extractor. Each modality yields a posterior correspondence map, p^{img} and p^{geo}, which are fused via the proposed *"Match-then-Fuse"* probabilistic module into a unified posterior p^{fuse}, from which the final rigid transformation is estimated.

Figure 2: Shows the complete C-GenReg architecture with dual branches (Generated-RGB and Geometric), probabilistic fusion module, and information flow from point clouds through World/Vision Foundation Models to final transformation estimation.

Limitations

- Computationally expensive due to World Foundation Model requiring ~507s per registration (though can be reduced 72x with model distillation)
- Relies on availability of coarse scene context (indoor/outdoor) for prompt guidance, though minimal prompts still work reasonably well
- Performance depends on quality of depth map generation from LiDAR for outdoor scenarios using virtual camera projection
- Limited to rigid transformations and does not handle non-rigid deformations or dynamic scenes